# William Zhou

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## EDUCATION

**North Carolina State University** — **Raleigh, North Carolina, USA**
*Bachelor of Science, Electrical and Computer Engineering* — *Expected May 2028*
- GPA: 3.97/4.00
- Relevant Coursework: Statistical Inference, Electric Circuits, Computer System Programming, Digital Logic Design
- High School Dual Enrollment: Multivariable Calculus, Linear Algebra, Differential Equations, Discrete Mathematics

## EXPERIENCES

### Research Intern
*Biomechatronics Lab, North Carolina State University* — *June 2024 - August 2024*
- Designed and fabricated hardware in Fusion 360 for a novel prosthetic hand with tactile feedback and robotic arm transportation system
- Developed user control interfaces in C++ using Arduino ESP32 microcontrollers and PlatformIO, enabling intuitive robotic arm manipulation
- Authored research abstract and poster on prosthetic robotic arm technology, accepted for presentation at IEEE International Conference on Advanced Intelligent Mechatronics (AIM) 2024

### Founder, Board Member
*Triangle Aerial Drone Association* — *September 2022 - Present*
- Dedicated to promoting drone technology in the North Carolina Research Triangle Park region.
- Mentored and taught two drone robotics teams to top 3 national placements at National REC Aerial Drone Competition
- Developed comprehensive technical resource guide analyzing competition drone source code, reverse-engineering autonomy stacks and implementing optimal navigation methods

### Engineering Communication Intern
*College of Engineering, North Carolina State University* — *September 2025 - Present*
- Responsible for brainstorming, filming and editing videos for the College of Engineering social media
- Collaborated with faculty, students, and other college departments to produce promotional content showcasing engineering programs and research initiatives
- Utilized Adobe Premiere Pro to produce high-quality video for university social media outreach on quick turnarounds

### Research Intern
*Acoustofludic Lab, Duke University* — *June 2023 - August 2023*
- Advanced the development of an innovative acoustic radiation force-based technique for precise drug delivery into microfluidic droplets.
- Optimized fluid circuit design using AutoCAD to determine the ideal convergence angle for fluid streams, enhancing droplet formation efficiency for acoustofluidic beaming applications.
- Designed and refined figures for poster presentations, meticulously documented iteration changes, and contributed to testing microfluidic devices using oscilloscopes and high-speed microscopy.

## CERTIFICATIONS/AWARDS

**FAA Part 107 Commercial Remote Pilot, EASA A1-A3 Remote Pilot, USACO Silver**

## SKILLS

**Programming** Python, C/C++, Git, LaTeX
**Platform and Tools** Betaflight, Ardupilot, KiCAD, Fusion360, Orcaslicer, Excel, Photoshop, Premiere Pro, Asana Enterprise
**Technical Skills** Oscilliscope, Function Generator, 3D-Printing, Soldering

## PROJECTS

- **First Persion View Drones**
  - Designed and built numerous first-person view drone systems, integrating custom and off-the-shelf components
  - Utilized Fusion360 and KiCAD to design custom components and mounts, expanding system capabilities
  - Optimized Electronic Speed Controllers and Flight Controllers through advanced PID tuning, implementing dynamic/low-pass filtering, and configuring UART/RC protocols to maximize system stability and responsiveness
  - Streamlined electronics and hardware selection to ensure compatibility and cost efficiency, reducing system costs by 30% compared to off-the-shelf solutions
- **NCSU Aerial Robotics Club**
  - Converted a manually operated drone into a dual-function system enabling autonomous waypoint navigation and manual payload release tests
  - Reprogrammed flight controller by flashing modified Ardupilot firmware for implementation of payload drop test drone
  - Redesigned and implemented a servo-actuated payload release system supporting both autonomous and manual control modes for testing